

MSc Data Science Project

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Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Exploring Renewable Energy and Energy Efficiency Impacts on
Sectoral Economic Growth in Low- and Middle-Income Countries.

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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at [Assessment Offences and Academic Misconduct](#) and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

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5.1 Tools, Libraries, and Environment

To conduct this comprehensive analysis, I worked extensively in **Python**, alternating between **Jupyter Notebook** for iterative local development and **Google Colab** for resource-intensive tasks such as rendering large heatmaps and geospatial plots. This hybrid setup allowed me to optimize both speed and flexibility depending on the computational needs of each task.

I employed a wide suite of powerful Python libraries, each carefully selected based on their capabilities in data science and machine learning:

- **Pandas** and **NumPy** served as my core tools for managing and transforming large-scale, multi-year, multi-indicator time-series data. I used them for reshaping, merging, missing value handling, and statistical computations.
- **Matplotlib** and **Seaborn** were used for crafting an extensive range of visualizations — including box plots, violin plots, multi-type time series trends, histograms, correlation heatmaps, PCA scree plots, and clustering diagrams. I relied on Seaborn's expressive color palettes (e.g., "coolwarm", "Set2") for visual clarity.
- **Scikit-learn** (sklearn) was the backbone of my **machine learning workflow**. I used it for **data standardization (StandardScaler)**, **dimensionality reduction (PCA)**, **unsupervised learning (KMeans clustering)**, and **internal validation metrics (silhouette score)**.
- For **geospatial visualization**, I turned to **GeoPandas** in combination with **Natural Earth shapefiles** (ne_110m_admin_0_countries.shp). This enabled me to map country clusters geographically and observe regional patterns that transcended income level boundaries.

Regarding data sources, I used **two distinct datasets**:

- The **primary dataset** was a World Bank-derived CSV file containing 91 economic, energy, and environmental indicators per country from **1989 to 2023**. This dataset was pre-structured into columns such as "Series Name", "Country Code", and yearly values (e.g., "1990 [YR1990]", "2023 [YR2023]"), which I then cleaned and transformed for analysis.
- The **income classification dataset** was retrieved from the official World Bank classification portal via the URL:
<https://databank.worldbank.org/data/download/site-content/CLASS.xlsx>.
This Excel sheet contained yearly income categorizations such as "Low income", "Lower middle income", and "Upper middle income" for all recognized countries.

In total, this environment and toolchain enabled me to maintain a consistent, modular, and highly replicable analytical pipeline from preprocessing through to final clustering and visualization stages.

5.2 Data Preprocessing

To prepare the dataset for rigorous, income-level-based analysis, I undertook a series of highly structured preprocessing steps. I combined technical precision with logical checks to ensure the dataset was reliable, standardized, and robust for all downstream statistical and machine learning procedures.

I began by importing essential libraries — NumPy, Pandas, and Matplotlib — to facilitate numerical operations, dataframe manipulation, and initial inspections. I loaded the raw dataset from the World Bank repository using the `.read_csv()` function and immediately addressed the issue of placeholder values ("..") by replacing them with `pd.NA`, thereby marking them as true missing values in Pandas.

Next, I isolated only the year-specific columns starting from the 5th column onwards using `df.columns[4:]`, and applied `pd.to_numeric()` with `errors='coerce'` to convert all these values into float-compatible formats. This ensured consistent numerical interpretation of values across time, coercing any remaining anomalies into `NaN`.

To impute these missing values meaningfully, I applied **group-wise forward and backward filling** — first by grouping the dataframe by ["Country Name", "Series Name"] and then transforming the year columns using a chained `.bfill().ffill()` method. This technique maintained the integrity of time-series continuity, ensuring that missing values within a country's series were filled only using that same country-series context.

While working on post-fill quality, I noticed some anomalous values in the 2023 column. To avoid inflated values corrupting my trend analysis, I inserted a rule-based filter where any 2023 value greater than 1.5× of its corresponding 2022 value was replaced with the 2022 value. This step was implemented using the `.where()` method:

```
df["2023 [YR2023]"] = df["2023 [YR2023]"].where(df["2023 [YR2023]"] < df["2022 [YR2022]"] * 1.5, df["2022 [YR2022]"])
```

Following this, any remaining `NaN` values were imputed using the **row-wise mean of each country-series pair**, implemented via `.apply(lambda row: row.fillna(row.mean()), axis=1)`. This strategy preserved the contextual distribution of each individual row rather than borrowing information across rows or columns.

To avoid distortions from sparsely populated series, I calculated a missing-value threshold set at **50% of the number of year columns**. Any row (i.e., country-indicator pair) with missing values exceeding this threshold was dropped entirely using `df.dropna(thresh=missing_threshold, axis=0)`, ensuring that only meaningful rows were retained for downstream analysis.

Once my base dataset was cleaned, I moved to merge it with the **World Bank's income classification dataset** for 2024–2025, sourced from:

<https://databank.worldbank.org/data/download/site-content/CLASS.xlsx>.

I used `pd.read_excel()` with `skiprows=3` and custom column names to ingest the sheet titled "**List of economies**", extracting only ["Country Name", "Country Code", "Income Level"] for the merge. Using an inner join on both "Country Name" and "Country Code" allowed me to accurately map each country to its respective income group.

After the merge, I identified unmatched records — i.e., countries with NaN in "Income Level" — using logical filtering. Many of these were **regions or aggregates**, not actual countries, such as "World", "Euro area", or "Sub-Saharan Africa". I compiled a comprehensive list of these aggregate entries and removed them using `~df["Country Name"].isin(regions_to_remove)`. This filtered the dataset to country-level rows only.

Still, a handful of actual countries remained unmatched due to naming mismatches or edge-case classifications. I reviewed these manually and constructed a mapping dictionary of corrected "Income Level" values using domain knowledge and World Bank classification updates. For instance:

```
manual_income_levels = {
    "Cote d'Ivoire": "Lower middle income",
    "Sao Tome and Principe": "Lower middle income",
    "Turkiye": "Upper middle income",
    "Viet Nam": "Lower middle income",
    "Afghanistan": "Low income",
    "Venezuela, RB": "Not classified",
    ...
}
```

I applied this mapping using `.map()` and `.fillna()` to ensure a fully populated "Income Level" column. Next, I **explicitly excluded** high-income countries and "Not classified" entries using a final logical filter. This brought the dataset down to only three groups: **Low income**, **Lower middle income**, and **Upper middle income**.

To validate data quality, I conducted an assertive check for duplicates using `.duplicated().sum()` and confirmed that no duplicates remained. Additionally, I verified that the "Income Level" column contained only the three intended classifications via an assertion check:

```
assert all(income in valid_income_levels for income in unique_income_levels)
```

As a final step, I incorporated expert recommendation updates. For instance, I removed "Bulgaria", "Palau", and "Russian Federation" manually, as these countries had just transitioned to the high-income bracket (World Bank, 2025). This filtering ensured temporal consistency between our data and the latest World Bank classification.

I then saved the fully processed dataset to disk using `.to_csv()`, ensuring reproducibility. From this point forward, the resulting dataset was clean, structured, and enriched with

income group metadata — making it ready for exploratory, statistical, and machine learning-based analyses

5.3 Exploratory Data Analysis (EDA)

After preparing the dataset, I transitioned into exploratory analysis to derive meaningful insights. My EDA process was comprehensive and multi-layered, spanning descriptive statistics, visual exploration, time-series tracking, and correlation analysis. I ran the analysis using a combination of **Pandas**, **Seaborn**, **Matplotlib**, and **NumPy**, all within the Jupyter environment.

Thematic Correlation Analysis: Linking All Four Research Dimensions

To directly address my research question, I compiled four indicator blocks:

1. **Renewable energy adoption**
2. **Energy efficiency**
3. **Sectoral economic growth**
4. **Emissions and carbon impact**

I manually curated lists of relevant series under each block (e.g., *Renewable energy consumption (% of total final energy consumption)*, *Energy use per \$1,000 GDP*, *Services value added growth*, *Total greenhouse gas emissions excluding LULUCF*, etc.). These were combined into one master list: `all_research_series`.

Sector-Based Correlation Analysis

Lastly, I explored how **CO₂ emissions, energy use, and renewables** influenced **sectoral growth** in **Agriculture, Industry, and Services**. I defined a dictionary `sector_series_dict` and looped over each income level and sector.

For each combination, I:

- Filtered the dataset by income and selected sector-relevant indicators
- Averaged year-wise values to derive `Series Avg`
- Created a country-series pivot table using `pivot_table()`
- Computed **three different correlation matrices** — Pearson, Spearman, and Kendall — for each sector
- Visualized each matrix using `sns.heatmap()` with customized formatting

This yielded a total of **27 heatmaps** (3 sectors × 3 income levels × 3 correlation types). Combinations with insufficient data (`pivot_df.shape[0] < 3`) were automatically skipped to preserve statistical integrity. The multi-method approach helped me triangulate sectoral insights and reduce the risk of over-reliance on a single correlation paradigm.

This exhaustive EDA served not only as the diagnostic phase of my project but also laid the groundwork for dimensionality reduction and cluster-based grouping in later stages. It helped me identify structural similarities across countries, changes over time, and thematic patterns across income classes.

5.4 Principal Component Analysis (PCA)

Once I had established thematic patterns from EDA, I transitioned into **dimensionality reduction** using **Principal Component Analysis (PCA)** to reveal latent structures and compress multivariate relationships across income levels. I handled PCA separately for **Low**, **Lower-middle**, and **Upper-middle** income countries, treating them as **independent data universes**.

To begin, I manually compiled a list of 17 indicators by merging four blocks—**renewable energy**, **energy efficiency**, **sectoral economic growth**, and **CO₂ emissions**—that mapped to the four core themes of my research question. I selected these series to capture a holistic yet compact representation of energy-growth-emission interdependencies.

For each income group:

- I filtered the master dataset to retain only countries from the target income level and only rows where Series Name belonged to selected_series.
- Next, I **converted all year columns to numeric**, dropping any parsing-related issues using `errors='coerce'`, ensuring every series was analysis-ready.
- I computed a **row-wise mean across all years** to derive a time-agnostic column called "Series Avg" for every indicator-country pair, using `.mean(axis=1)`.
- Using `pivot_table()`, I transformed the dataframe so that each **row represented a country**, and each **column a research-relevant indicator**. I excluded countries that had missing data across any of the selected series by applying `.dropna()`.

Then, I applied **standardization** using `StandardScaler()` from scikit-learn to ensure each series was centered and scaled (mean=0, variance=1). This step was crucial to prevent high-magnitude indicators from dominating the PCA components.

I fit PCA on the scaled matrix with `PCA().fit_transform()` to reduce the dataset into a new **orthogonal coordinate system**. From this:

- I captured **explained variance ratios** of each principal component
- I generated a **scree plot** showing the cumulative variance explained, helping visualize how much information could be preserved using only two components

Finally, I constructed a **two-dimensional PCA scatter plot**, showing countries in the new PC1–PC2 space using `sns.scatterplot()`. I called `plot_pca_with_labels()` to annotate each point with its country name, enhancing interpretability.

After generating PCA plots, I aimed to go beyond visual clustering by **quantifying inter-country similarity**. For this, I computed a **Euclidean distance matrix** between countries

using the original PCA input space (i.e., standardized Series Avg vectors). I implemented this through `pdist()` and converted it into a squareform matrix.

To enhance visual clarity and interactivity, I created an **interactive Plotly heatmap** using `px.imshow()`, which displayed country-wise distances inside each heatmap cell and allowed zooming and panning. These plots were saved in HTML format for documentation and easy sharing.

Through this, I uncovered patterns of intra-income group heterogeneity — for instance, while some Low-income countries clustered tightly (e.g., Chad and Congo Dem. Rep.), others like Ethiopia or Afghanistan appeared distant.

This helped me realize that not all countries in the same income level behave the same way, which challenges the assumptions behind income-level grouping and justifies my use of PCA and clustering.

5.5 Clustering and Profile Analysis

Building on the PCA foundation, I conducted **unsupervised clustering** using the **KMeans algorithm** on the PCA coordinates for each income group. I designed a function `run_kmeans_clustering()` that:

- Accepted PCA coordinates and maximum `k` (default 12)
- Iteratively ran KMeans from **`k = 2` to optimal**, storing **inertia** and **silhouette scores**
- Chose optimal `k` using the **maximum silhouette score**, then fit a final KMeans model to label each country

I visualized the **elbow curve** to show inertia reduction per cluster count, helping support the chosen `k`. Then, I plotted the final cluster results in the PC1–PC2 space using **cluster-wise scatter plots** with text annotations for each country. These helped visually evaluate clustering separation and density.

To understand what defined each cluster, I profiled them using the original (non-PCA) indicators:

- I reloaded the original indicators per country using a filter on Income Level and countries used in PCA
- I grouped by cluster label and calculated **cluster-wise mean values** for all indicators
- I normalized those averages using **min–max scaling**, and generated **radar-like line plots** to contrast profiles across clusters visually

Next, I implemented **geographic mapping of clusters** using GeoPandas. I downloaded the **Natural Earth shapefile** for country boundaries and merged it with my cluster data. I corrected known naming mismatches between World Bank names and shapefile names using a manual `rename_map` dictionary. Then:

- I plotted each country shaded by cluster assignment

- I annotated maps with **country names using centroid coordinates**, keeping label sizes readable
- I repeated this for all income levels, generating three separate maps

Finally, to facilitate comparative interpretation, I wrote a composite function `plot_all_income_levels_geoclusters()` that:

- Mapped the **clustered countries side-by-side** using `subplots(1, 3)`
- Harmonized plotting scale, colouring, and legend handling
- Created a **visual dashboard of global clustering outcomes** by income level

This multifaceted clustering approach allowed me to:

- Identify **latent country groupings** beyond income level
- Profile each group in terms of indicator performance
- Visualize global **geospatial patterns** in energy-growth-emission profiles

Together, PCA and KMeans uncovered **hidden structures** that would have been lost in standard analysis, making them essential for the interpretative depth of this project.

5.6 Panel Regression Methodology

To rigorously examine the sectoral economic growth drivers across income groups, I employed panel data regression techniques using both Fixed Effects and Random Effects models. My goal was to quantify the influence of selected energy, emissions, and infrastructure indicators on the growth of agriculture, industry, and services sectors.

Library and Environment Setup

I used Python in a Jupyter environment, leveraging the `pandas` and `numpy` libraries for data manipulation, and `linearmodels` for estimating panel regressions. Additionally, I imported `statsmodels.api.add_constant` for including intercept terms, and `scipy.stats` for executing the Hausman test.

```
from linearmodels.panel import PanelOLS, RandomEffects
from statsmodels.tools.tools import add_constant
from scipy import stats
```

Selection of Target Variables

I focused on three key sectoral growth indicators from the World Bank:

- **Agriculture, forestry, and fishing, value added (annual % growth)**
- **Industry (including construction), value added (annual % growth)**

- **Services, value added (annual % growth)**

Each served as a dependent variable in a separate model. For each sector, I executed regressions individually for **low-income**, **lower-middle-income**, and **upper-middle-income** countries.

Independent Variable Selection

I selected the following five predictors based on domain knowledge, EDA, and correlation analysis:

1. Renewable energy consumption (% of total final energy consumption)
2. Energy intensity level of primary energy (MJ/\$2017 PPP GDP)
3. Access to electricity (% of population)
4. GDP per unit of energy use (PPP \$ per kg of oil equivalent)
5. Carbon dioxide (CO₂) emissions excluding LULUCF per capita (t CO₂e/capita)

These features were consistently used across all sectoral models. I later experimented with adding GDP growth (annual %) and PM_{2.5} air pollution based on their potential relevance and availability.

Data Preparation

I defined a reusable function `prepare_panel_data()` that extracted and reshaped the panel dataset for a given income level and dependent variable. This function:

- Filtered countries by income group
- Selected only relevant series names
- Melted the dataset from wide to long format
- Pivoted it to ensure each row represented a (country, year) pair
- Dropped missing values
- Converted year to integer

```
df_long = df_filtered.melt(id_vars=['Country Name', 'Series Name'],
var_name='Year', value_name='Value')
df_panel = df_long.pivot_table(index=['Country Name', 'Year'],
columns='Series Name', values='Value').reset_index()
```

I renamed columns for clarity and ensured the dataset had no nulls before regression.

Regression Execution

For each prepared dataset (e.g., `agri_df_low`), I used `PanelOLS` to fit the **Fixed Effects** model and `RandomEffects` to estimate the **Random Effects** model. Both models were fitted with **robust standard errors** to ensure valid inference under heteroskedasticity.

```
fe_model = PanelOLS(y, X,  
entity_effects=True).fit(cov_type='robust')  
re_model = RandomEffects(y, X).fit(cov_type='robust')
```

Hausman Test and Model Choice

To determine whether the Fixed or Random Effects model was more appropriate, I implemented the **Hausman test** manually. I computed the difference in coefficients and their covariance matrices to derive the test statistic and p-value.

```
b_diff = fe_results.params - re_results.params  
cov_diff = fe_results.cov - re_results.cov  
stat = b_diff.T @ np.linalg.inv(cov_diff) @ b_diff
```

If the p-value was below 0.05, I preferred the Fixed Effects model; otherwise, I used Random Effects. The model selection logic and summaries were printed for transparency.

Model Evaluation and Visualization

After model fitting, I extracted the predicted values and added them back to the original dataset for comparison. I then plotted the **actual vs predicted growth** for one representative country per income group, using `matplotlib`.

```
plt.plot(df_sample['year'], df_sample['actual'], label='Actual',  
marker='o')  
plt.plot(df_sample['year'], df_sample['predicted'],  
label='Predicted', marker='x')
```

PCA-based Validation

As a final methodological step, I conducted robustness validation by restricting the dataset to countries selected through prior Principal Component Analysis (PCA). These countries were identified as structurally representative within their income group based on energy-emission-growth profiles.

To implement this:

- I loaded PCA coordinate files for each income group.

- Then I filtered the main dataset to retain only those countries present in the PCA coordinate set.
- Using the filtered subset, I reconstructed the panel datasets and reran the same fixed and random effects models for a single sector and income group as a **targeted diagnostic test**, keeping all variable selections and model specifications unchanged.

6. Results

6.1 Correlation Heatmap for Sectoral Economic Growth

I examined the relationships between agriculture sector growth and environmental-energy indicators across low-income countries using Pearson, Spearman, and Kendall correlation methods. Across all approaches, there was a consistent negative correlation between agricultural value added and CO₂ emissions per capita (Pearson: -0.55, Spearman: -0.39, Kendall: -0.43), suggesting that higher emissions may be linked to reduced agricultural growth. Renewable electricity output also showed a strong negative association (Pearson: -0.56, Spearman: -0.61, Kendall: -0.33). In contrast, electricity production from renewable sources (excluding hydro) exhibited positive correlations, particularly under Spearman (0.48), indicating a potential supportive role of renewables. Combustible renewables and waste displayed a modest positive relationship across all methods.

For the industry sector in the same income group, industry growth was strongly and positively correlated with energy use per \$1,000 GDP (Pearson: 0.81), carbon intensity of GDP (0.42), and total greenhouse gas emissions (0.46), suggesting that energy-intensive and emission-heavy activities might be driving industrial output. The Spearman and Kendall results reinforced these associations, particularly in relation to energy use and carbon indicators. At the same time, GDP per unit of energy use had consistently negative correlations across Pearson (-0.54), Spearman (-0.46), and Kendall (-0.24), indicating lower energy efficiency in rapidly growing industries. The relationship between manufacturing growth and overall industry growth was also strong (Pearson: 0.74, Spearman: 0.96, Kendall: 0.90), reinforcing their interdependence.

In the services sector, energy use per \$1,000 GDP maintained a strong positive correlation with growth (Pearson: 0.82, Spearman: 0.68, Kendall: 0.52), while GDP per unit of energy use displayed a strong negative relationship (Pearson: -0.72, Spearman: -0.68, Kendall: -0.52). This indicated that higher energy intensity may be contributing to services expansion. Additionally, renewable energy consumption and renewable electricity output showed moderate positive correlations across all methods. On the other hand, fossil fuel energy consumption was negatively associated with services growth (Pearson: -0.63), particularly under Spearman (-0.64) and Kendall (-0.62), highlighting the influence of cleaner energy sources on sectoral development.

Shifting to lower middle-income countries, the correlations for the agriculture sector were generally weaker. Electricity production from renewable sources showed a consistent negative association with agriculture (Pearson: -0.29, Spearman: -0.23, Kendall: -0.17), indicating limited integration of renewables in this area. Other indicators mostly demonstrated weak or negligible correlations. Energy use per \$1,000 GDP and energy intensity had weak negative relationships, while GDP per unit of energy use showed a small

positive correlation (Pearson: 0.08, Kendall: 0.12). Greenhouse gas emissions, however, had slight positive correlations in Spearman (0.25) and Kendall (0.18), though not strongly influential.

In the industry sector, the results were mixed. CO₂ emissions excluding LULUCF consistently had a moderate negative correlation with industrial growth (Pearson: -0.46, Spearman: -0.44, Kendall: -0.31), suggesting that emissions may hinder expansion. Carbon intensity and fossil fuel energy use followed similar weak to moderate negative trends. Notably, combustible renewables and renewable energy consumption had positive correlations (Pearson: 0.38 and 0.35), pointing to a potentially supportive role for renewables. The connection between manufacturing and industry growth remained moderately strong across Pearson (0.62), Spearman (0.76), and Kendall (0.60).

As for the services sector in this income group, the associations were relatively weak overall. Greenhouse gas emissions excluding LULUCF showed moderate positive correlations with services growth (Pearson: 0.27, Spearman: 0.41, Kendall: 0.30), implying that emissions might indirectly accompany sectoral expansion. Combustible renewables and renewable energy consumption showed minor positive correlations, while most energy efficiency metrics—including GDP per unit of energy use and energy intensity—exhibited weak or near-zero relationships.

Among upper middle-income countries, the agriculture sector showed a more nuanced pattern. CO₂ emissions excluding LULUCF had a consistent negative correlation with agricultural growth (Pearson: -0.31, Spearman: -0.35, Kendall: -0.23), indicating environmental pressure on productivity. On the other hand, renewable energy consumption (Pearson: 0.29) and renewable electricity output (Pearson: 0.23) had moderate positive associations, suggesting benefits from renewable adoption. Fossil fuel energy consumption was negatively correlated across all methods, while greenhouse gas emissions (excluding LULUCF) had small to moderate positive correlations (Pearson: 0.28), adding complexity to the emissions-agriculture relationship.

In the industrial sector, correlations were mostly weak. Fossil fuel energy consumption had a slight positive relationship with industry growth (Pearson: 0.21), while CO₂ emissions and greenhouse gases demonstrated weak negative associations across all methods. Renewable energy indicators—including renewable consumption and electricity output—had consistent but weak negative correlations (e.g., Pearson: -0.20 and -0.07), suggesting limited renewable integration. Meanwhile, manufacturing growth was modestly related to overall industrial growth (Pearson: 0.21, Spearman: 0.54, Kendall: 0.42).

Finally, in the services sector, relationships were largely inconsistent and weak. Total CO₂ emissions showed a moderate positive Pearson correlation (0.42), but the relationship faded in Spearman and Kendall. Most energy-related indicators—such as energy intensity, GDP per unit of energy use, and renewable electricity output—had near-zero correlations. Renewable energy consumption showed almost no association with services growth (Pearson: -0.004). Greenhouse gas emissions excluding LULUCF had a moderately positive Pearson value (0.41) but turned negative in the other methods, illustrating the complex and variable nature of emissions impacts on the services sector across countries.

6.2 PCA and Euclidean Distance Analysis Across Income Groups

To explore the structural similarities among countries within each income group based on energy and environmental indicators, I applied **Principal Component Analysis (PCA)** to reduce dimensionality and then computed **Euclidean distances** based on the resulting coordinates. This method allowed me to assess inter-country similarity in a lower-dimensional space while preserving as much information as possible.

Low-Income Countries

For low-income countries, the first **two principal components** captured a substantial portion of the variance in the dataset, with the **cumulative explained variance** reaching **92.5%** at PC4 and exceeding **98%** at PC5. The **PCA scatter plot** revealed notable spatial separations. For instance, **Sudan** (PC1 = 4.99, PC2 = -1.80) appeared as a clear outlier compared to countries like **Ethiopia** (PC1 = -2.90, PC2 = -2.23) and **Mozambique** (PC1 = -2.80, PC2 = -1.55), suggesting significant divergence in their energy-emission profiles.

Using the PCA-transformed coordinates, I calculated pairwise **Euclidean distances** between countries, visualized via a heatmap. The **smallest distance** was observed between **Congo, Dem. Rep.** and **Zambia** (Distance = 37), indicating strong similarity. In contrast, **Mozambique** and **Sudan** showed the **largest distance** (Distance = 433), reflecting substantial dissimilarity in their underlying variables. Other closely grouped countries included **Ethiopia and Mozambique** (Distance = 93), while **Niger and Sudan** (Distance = 108) also appeared relatively similar.

Lower Middle-Income Countries

In the lower middle-income group, the **first six PCA components** explained approximately **88% of the cumulative variance**, crossing **95%** by the eighth component. The **PCA scatter plot** showed widespread dispersion, with **India** (PC1 = 1.30, PC2 = 5.58) and **Ukraine** (PC1 = 6.54, PC2 = -3.48) positioned at extremes. This highlighted significant diversity in the energy-emission characteristics across this income group.

The corresponding **Euclidean distance matrix** revealed the **closest country pair** to be **Sri Lanka and Angola** (Distance = 38.81), followed closely by **Cameroon and Angola** (Distance = 57.39), and **Nicaragua and Cote d'Ivoire** (Distance = 33.65), suggesting these nations have nearly identical environmental-energy profiles after PCA reduction. Conversely, **India and Bangladesh** (Distance = 2714.88), and **India and Congo, Rep.** (Distance = 2884.27) formed some of the **most distant pairs**, pointing to distinct energy-emission behaviors and levels of industrial development. These results underscore a broad spectrum of structural patterns within this income group.

Upper Middle-Income Countries

Among upper middle-income countries, the **first six principal components** explained over **90% of the cumulative variance**, reaching nearly **100%** by the twelfth. In the **PCA scatter plot**, countries like **Kazakhstan** (PC1 = 5.67, PC2 = 2.21), **South Africa** (PC1 = 4.58, PC2 =

1.93), and **Paraguay** (PC1 = -4.30, PC2 = 2.23) were visibly distant from the central cluster, revealing regional extremes in energy and emission metrics, with **Gabon** (PC2 = 6.29) being a major outlier in PC2.

From the **Euclidean distance matrix**, the **closest pair** was **Thailand and Turkiye** (Distance = 84.63), followed by **Namibia and Suriname** (Distance = 55.03), and **Dominican Republic and Suriname** (Distance = 47.25), indicating shared characteristics in reduced PCA space. On the other hand, **Brazil** showed the **largest distances** from many countries, such as **Brazil and Albania** (Distance = 1082.91) and **Brazil and Botswana** (Distance = 1084.13), reflecting significant divergence in energy-related indicators. This variation highlights how regional energy policies, resource structures, and development priorities shape the clustering of nations.

6.3 Clustering and Geographic Patterns

Low Income Countries

Optimal Number of Clusters: The elbow plot clearly indicated an inflection at **k=5**, confirming that five clusters capture the inherent structure of low-income country data.

PCA-Based Clustering: The scatter plot using the first two principal components revealed well-separated clusters, demonstrating a meaningful distribution of countries in the reduced feature space. Each cluster represents distinct socio-environmental trajectories, with countries like *Sudan* and *Congo, Dem. Rep.* occupying distant positions, indicating unique profiles.

Geographic Cluster Distribution: The spatial plot highlighted that clustered countries were not strictly region-bound. For instance, *Zambia* and *Togo* belonged to the same cluster despite regional separation, reinforcing the notion that clustering was based on developmental indicators rather than geography.

Cluster Profile Trends: The temporal indicator trend showed that **Cluster 0** maintained the highest normalized values consistently post-2005, indicating sustained performance. Meanwhile, **Cluster 3 and 4** exhibited earlier peaks followed by declines or stagnation, suggesting episodic improvements possibly affected by instability or policy reversals.

Lower Middle Income Countries

Optimal Number of Clusters: The elbow method showed a sharp drop in inertia at **k=2**, suggesting a binary structure. This split was consistent with two development pathways.

PCA-Based Clustering: The PCA scatter plot showed a clear bifurcation. **Cluster 0** encompassed the majority of countries, while **Cluster 1** grouped outliers like *India*, *Iran*, *Mongolia*, *Ukraine*, and *Uzbekistan*, reflecting diverging indicator patterns — likely due to energy production or CO₂ profiles.

Geographic Patterns: The map revealed that **Cluster 1** countries were geographically dispersed across South Asia, Central Asia, and Eastern Europe. Despite regional variety, they

shared common developmental patterns, possibly influenced by higher energy consumption, industrial output, or policy alignment.

Cluster Profile Analysis: The cluster profile plot revealed a sharp transition around **2002–2003**, where most countries moved from **Cluster 0** to **Cluster 1**. This aligns with global economic shifts and the impact of the **Millennium Development Goals (MDGs)**. The consistent high performance of Cluster 1 post-2003 suggests long-term gains in energy access, emissions management, or GDP-energy efficiency.

Upper Middle Income Countries

Optimal Number of Clusters: The elbow method suggested **k=4** clusters as the optimal partition, capturing nuanced groupings beyond just economic output.

PCA-Based Clustering: Countries like *Gabon*, *Paraguay*, and *Guatemala* formed unique clusters (e.g., Cluster 3), whereas others like *South Africa* and *Kazakhstan* grouped together (Cluster 2), possibly due to industrial similarities or emissions intensities.

Geographic Cluster Mapping: The map reflected a mix of regional clustering and outlier behavior. *South American countries* largely fell into **Cluster 0**, while *South Africa* and *Kazakhstan* clustered together in **Cluster 2**, reflecting potential alignment in energy-intense industrial sectors.

Temporal Profiles: Cluster 0, maintaining the **highest normalized scores throughout**, indicates stable leaders in key development indicators. Cluster 1, while initially close to Cluster 0, declined steadily after 2000. Cluster 3 showed a late rise in the 2010s, suggesting policy or technological shifts. Cluster 2, although present, maintained the lowest values, likely due to specific energy or emissions inefficiencies.

7. Analysis & Discussion

Models trained on globally inclusive datasets (within income level) demonstrate stronger generalizability to all member countries. However, restricting the training set to PCA-included countries leads to models that are highly tuned to those specific country patterns, often sacrificing out-of-sample performance. This trade-off must be carefully considered depending on whether interpretability or predictive accuracy across all countries is prioritized.

8. Conclusion

9. References

10. Appendices